Name: **Numbers Test** **Date:** 24/09/2020

1. Place the first 10 natural numbers on the number line below.

1 2 3 4 5 6 7 8 9 10

2. List the first ten *odd* natural numbers.

1 3 5 7 9 11 13 15 17 19

3. List the first ten *even* natural numbers.

2 4 6 8 10 12 14 16 18 20

4. Using these four digits: 2, 9, 3, 5

(a) What is the *largest* four-digit number that can be made?

9532

(b) What is the *smallest* four-digit number that can be made?

2359

(c) Calculate the *difference* between the numbers from part **(a)** and **(b)**?

9532
2359
7173

good job.

5. **(a)** Write the following list of numbers in *ascending* order:
604, 406, 46, 0, 64

0, 46, 64, 406, 604

(b) Write the following list of numbers in *descending* order:
210, 85, 219, 26, 401, 79, 151

401, 219, 210, 151, 85, 79, 26

6. The number 12 has three pairs of *factors*: 1 x 12, 2 x 6 and 3 x 4.

(a) What are the pairs of factors of 18?

(b) What are the pairs of factors of 36?

23%

Try and answer more questions and I bet you'll do better next time!

7. List all the factors of each of these numbers:

(a) 6

2 × 3

(b) 15

(c) 28

8. Find the *highest common factor (HCF)* of these pairs of numbers:

(a) 12 and 16

(b) 15 and 25

9. Complete the list of the first eight *prime* numbers:

2, ____ , 5, ____ , ____ , ____ , ____ , ____

10. Write down the *prime factors* of:

(a) 12

(b) 16

(c) 35

11. Work out each of these numbers:

(a) 4^2

(b) 5^2

(c) 2^3

12. Work out each of these numbers:

(a) $\sqrt{9}$

(b) $\sqrt{36}$

(c) $\sqrt{49}$

13. Many maths problems have multiple steps to do, so it's important to do them in the right order. Some people use the term **BIMDAS** to remember the order.

BIMDAS:
1. Brackets
2. Indices (= powers and square roots)
3. Multiplication
4. Division
5. Addition
6. Subtraction

Use **BIMDAS** to solve these sums:

(a) (4 + 7) x (5 - 2)

(b) 25 ÷ (8 - 3) + 2

(c) 3 x 4 + 3 x (10 - 4)

(d) $\sqrt{4} + \sqrt{25}$

(d) $\sqrt{16} + 5 \times 2^3$

14. Use the *array method* to multiply the following numbers:

(a) 9 x 7

	5	+ 2
5 + 4		

(b) 14 x 15

(c) 21 x 34

(d) 56 x 44

(e) 136 x 7

(f) 127 x 46

(g) 225 x 12

(h) 121 x 133

15. Have a good weekend!